

Generative AI Internship – Task 1

PROJECT: Prompt Engineering Showcase

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Introduction

Prompt engineering is the process of designing and improving instructions given to Artificial Intelligence models to generate more accurate, relevant, and useful responses. This document demonstrates 20 effective prompts across different domains by comparing basic prompts with optimized prompts and explaining the improvements made.

1. Coding Domain

Prompt 1

Basic Prompt:

Make a website.

Optimized Prompt:

Create a responsive portfolio website using HTML, CSS, and JavaScript. Include sections for About Me, Skills, Projects, and Contact. Use a clean modern design and explain the code structure.

Improvement:

Added technology requirements, design expectations, and output details.

Prompt 2

Basic Prompt:

Fix my code.

Optimized Prompt:

Analyze the following Python code, identify errors, explain the cause of each issue, and provide the corrected version with improvements.

Improvement:

Specified the programming language and requested explanation along with solution.

Prompt 3

Basic Prompt:

Create an app.

Optimized Prompt:

Create a simple task management application using Python. Include features like adding tasks, deleting tasks, and marking tasks as completed. Provide step-by-step implementation.

Improvement:

Defined features and expected development approach.

Prompt 4

Basic Prompt:

Explain Java.

Optimized Prompt:

Explain Java programming concepts for a beginner with examples of variables, loops, classes, and object-oriented programming.

Improvement:

Defined audience and required topics.

2. Writing Domain

Prompt 5**Basic Prompt:**

Write about AI.

Optimized Prompt:

Write a beginner-friendly 500-word article explaining Artificial Intelligence, its applications, benefits, and limitations with examples.

Improvement:

Added audience, length, and structure.

Prompt 6**Basic Prompt:**

Write an email.

Optimized Prompt:

Write a professional email requesting internship information from a company. Keep the tone polite and concise.

Improvement:

Specified purpose and tone.

Prompt 7**Basic Prompt:**

Create a resume.

Optimized Prompt:

Create a resume format for a computer science student applying for internships. Include skills, projects, education, and technical experience.

Improvement:

Defined target user and required sections.

Prompt 8

Basic Prompt:

Write a LinkedIn post.

Optimized Prompt:

Write a professional LinkedIn post announcing my Generative AI internship. Keep it engaging and include relevant hashtags.

Improvement:

Added platform, purpose, and style.

3. Summarization Domain

Prompt 9**Basic Prompt:**

Summarize this.

Optimized Prompt:

Summarize this article into 5 important bullet points highlighting the main ideas and conclusions.

Improvement:

Defined format and important points.

Prompt 10**Basic Prompt:**

Make notes.

Optimized Prompt:

Convert this chapter into easy-to-understand study notes with headings, key points, and examples.

Improvement:

Specified learning format.

Prompt 11**Basic Prompt:**

Explain this document.

Optimized Prompt:

Analyze this document and provide a simple explanation of the main concepts for a beginner.

Improvement:

Defined explanation level.

Prompt 12

Basic Prompt:

Summarize a video.

Optimized Prompt:

Summarize this video transcript into key points, important examples, and final conclusions.

Improvement:

Specified output requirements.

4. Data Analysis Domain

Prompt 13**Basic Prompt:**

Analyze data.

Optimized Prompt:

Analyze this sales dataset, identify trends, calculate important metrics, and provide suggestions for improving business performance.

Improvement:

Added analysis goals and expected results.

Prompt 14**Basic Prompt:**

Find patterns.

Optimized Prompt:

Analyze this dataset and identify important patterns, relationships, and unusual values.

Improvement:

Defined what patterns to search for.

Prompt 15**Basic Prompt:**

Create a report.

Optimized Prompt:

Create a data analysis report including observations, insights, charts, and recommendations.

Improvement:

Specified report structure.

Prompt 16**Basic Prompt:**

Explain data.

Optimized Prompt:

Explain this dataset in simple terms and describe what each column represents.

Improvement:

Defined explanation requirement.

5. Creative Domain

Prompt 17**Basic Prompt:**

Write a story.

Optimized Prompt:

Write a mystery story with a female detective as the main character, a suspenseful atmosphere, and an unexpected ending.

Improvement:

Added genre, character, and style.

Prompt 18**Basic Prompt:**

Give ideas.

Optimized Prompt:

Generate 10 creative project ideas for a computer science student interested in Artificial Intelligence.

Improvement:

Specified target and topic.

Prompt 19**Basic Prompt:**

Create a caption.

Optimized Prompt:

Create 10 creative Instagram captions for a cozy reading and lifestyle account.

Improvement:

Added context and audience.

Prompt 20**Basic Prompt:**

Suggest a name.

Optimized Prompt:

Suggest unique and memorable names for an AI-based application. Names should be short, professional, and brand-friendly.

Improvement:

Added naming criteria.

Reusable Prompt Template Library

Coding Template

Act as a software developer. Create [project name] using [technology]. Include [requirements]. Explain the solution step-by-step.

Learning Template

Teach me [topic] as a beginner. Explain concepts clearly with examples and practical applications.

Writing Template

Write a [type of content] about [topic]. Use a [tone] style and target [audience].

Analysis Template

Analyze this [data/document]. Identify key insights, patterns, and provide recommendations.

Conclusion

This task helped me understand the importance of prompt engineering in Generative AI. By creating structured prompts with clear instructions, context, and expected outputs, AI models can generate more accurate, relevant, and useful responses. This project improved my understanding of designing effective prompts for different real-world applications.

Tools used

- ChatGPT
- Claude
- Microsoft Word